MUNICIPAL CORPORATION OF GREATER MUMBAI (MCGM / BMC)

RED WARNING

TIER_2_SCREENING_SIMULATION

DISASTER MANAGEMENT CELL • EMERGENCY OPERATIONS EXECUTIVE DOSSIER

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EXECUTIVE SITUATION SUMMARY:

Multi-criteria hydrodynamic simulation evaluated for scenario **"Scenario A: Monsoon Downpour & Low-Lying Waterlogging"** across all 24 canonical administrative wards of Greater Mumbai over a **T+2h** forecast horizon. Topographical saucer basins and coastal backpressure were evaluated against Copernicus GLO-30 DEM terrain profiles.

SEVERE WARDS	HIGH RISK WARDS	CITYWIDE MEAN RISK	SCREENED POPULATION
1	11	54.9	~1588k
Score ≥ 80 / 100	Score 60 - 79 / 100	Normalised 0-100 scale	Low-lying contour buffers

1. KEY INTELLIGENCE FINDINGS

EVIDENCE-BASED SYNTHESIS

- Peak hazard susceptibility is concentrated in Ward G/North (Dadar / Dharavi / Mahim) at a score of 84.4/100.
- Top intervention priorities across Mumbai's 24 administrative wards: Ward G/North (84.4/100), Ward F/South (78.5/100), Ward F/North (78.5/100), Ward E (72.5/100), Ward L (69.7/100).
- Estimated population residing within modeled low-lying screening zones is ~15,88,100 across all wards.
- Critical infrastructure in proximity includes 7 tertiary hospitals and 12 major arterial transit segments.
- Overall municipal alert level is classified as RED_WARNING under applied parameters.

2. HIGHEST PRIORITY INTERVENTION WARDS

TOP 5 OF 24

RANK	WARD ID	WARD DESIGNATION	RISK SCORE	SEVERITY TIER	PRIMARY PHYSICAL TRIGGER
#1	Ward G/North	Ward G/North (Dadar / Dharavi / Mahim)	84.4 index (0-100)	SEVERE INUNDATION HAZARD	Low elevation + tidal backwater
#2	Ward F/South	Ward F/South (Parel / Sewri / Naigaon)	78.5 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#3	Ward F/North	Ward F/North (Matunga / Sion / Wadala)	78.5 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#4	Ward E	Ward E (Byculla / Nagpada / Mazgaon)	72.5 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#5	Ward L	Ward L (Kurla / Sakinaka / Asalpha)	69.7 index (0-100)	HIGH SINKING / CHOKE	Saucer depression ponding

3. POTENTIALLY EXPOSED MUNICIPAL INFRASTRUCTURE

CRITICAL ASSETS

Tertiary Municipal Hospitals — 7 assets

KEM Hospital, Sion Hospital, Nair Hospital, Cooper Hospital

Caveat: Located in spatial screening proximity (<500m); physical water ingress not confirmed.

Arterial Transit Corridors — 12 assets

Dr. Babasaheb Ambedkar Road, LBS Marg, Swami Vivekananda (SV) Road

Caveat: Road centerline intersects modeled basin buffer; traffic impassability subject to spot depth.

Copernicus GLO-30 Topographic Depressions — 6 assets

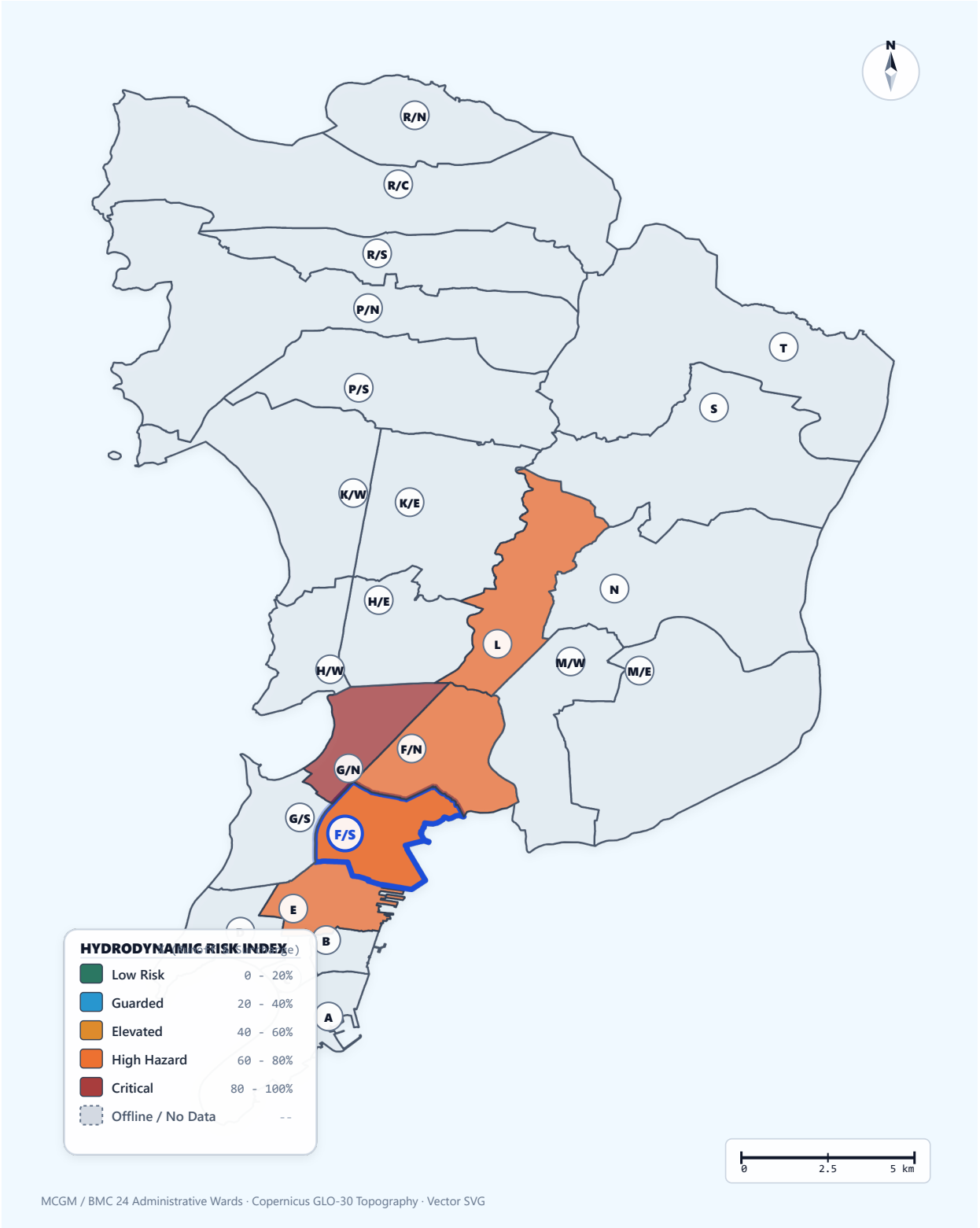
Hindmata Basin (4.5m ASL), Kurla Sump (5.2m ASL), Milan Subway (5.8m ASL)

Caveat: Documented elevation minima where gravity storm-water discharge reverses during high tide.

GREATER MUMBAI 24-WARD HYDRODYNAMIC CARTOGRAPHY

T+2h Simulation Horizon

FIGURE 1.1: MUNICIPAL CHOROPLETH RISK DISTRIBUTION & TOPOGRAPHIC SINK SCREENING



Cartographic Note: Wards are filled according to their calculated waterlogging and drainage surcharge score for the active scenario. Boundaries adhere to canonical BMC administrative boundaries (EPSG:4326). Blue boundary highlights selected ward focus. Shaded contours represent Copernicus GLO-30 DEM depression sinks.

COMPREHENSIVE 24-WARD MUNICIPAL EVALUATION MATRIX

24 BMC Administrative Wards

SECTION C: ADMINISTRATIVE WARD RANKINGS & POPULATION EXPOSURE

RANK	WARD	ADMINISTRATIVE WARD DESIGNATION	SCORE	SEVERITY	ELEV (M)	EXPOSED POP	ASSETS	PRIMARY TRIGGER
#1	G/North	Ward G/North (Dadar / Dharavi / Mahim)	84.4	SEVERE INUNDATION HAZARD	4.9m	~1,80,000	1	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#2	F/South	Ward F/South (Parel / Sewri / Naigaon)	78.5	HIGH SINKING / CHOKE	4.8m	~70,200	6	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#3	F/North	Ward F/North (Matunga / Sion / Wadala)	78.5	HIGH SINKING / CHOKE	5.1m	~95,400	3	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#4	E	Ward E (Byculla / Nagpada / Mazgaon)	72.5	HIGH SINKING / CHOKE	6.2m	~79,200	4	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#5	L	Ward L (Kurla / Sakinaka / Asalpha)	69.7	HIGH SINKING / CHOKE	6.9m	~1,62,000	5	Score = [RunoffExcess(0.0 mm/h over Cap 42.5) → 0.

* Population estimates derive from 2011 Census ward totals adjusted to 2026 BMC projections, intersected with low-lying saucer depression buffer zones.

WARD FOCUS: WARD F/SOUTH (PAREL / SEWRI / NAIGAON)

HIGH SINKING / CHOKE (78.5 / 100)

SECTION D: TOPOGRAPHIC PROFILE & CRITICAL ASSET EXPOSURE

ADMINISTRATIVE ZONE	MEAN GROUND ELEVATION	DEPRESSION HOTSPOTS	EXPOSED POPULATION
Zone II (South-Central Mumbai / Island City North)	4.8m	1	~70,200
Coastal Backwater Zone	Copernicus GLO-30 ASL	Chronic low points	of ~3,90,000 residents

TOPOGRAPHICAL SINKS & DRAINAGE OUTFALLS

Drainage Waterways & Basins: Hindmata-Parel Central Low-Lying Bowl & Sewri Outfalls, Arabian Sea Coastline

Depression Sinks: Hindmata - Parel Saucer Depression (6.7m)

Historical Chronic Waterlogging Points: Hindmata Flyover Underpass, Parel TT Circle, Dr. B.A. Road Lalbaug Depression, Chitra Cinema Stretch

POTENTIALLY EXPOSED MEDICAL & TRANSPORT INFRASTRUCTURE

HOSPITALS IN LOW-LYING BUFFER

- **BMC Public Health Department (Executive Health Officer Headquarters)** (Municipal Apex Public Health & Epidemiology Directorate)
- **King Edward Memorial (KEM) Hospital & Seth G.S. Medical College** (Municipal Tertiary Teaching Hospital & Medical College)
- **Sewri Group of TB Hospitals (GTB Hospital)** (Municipal Specialized Pulmonary & Tuberculosis Hospital)

MAJOR ARTERIALS IN INUNDATION CORRIDOR

- **Dr. Babasaheb Ambedkar Road (Lalbaug to Hindmata & Dadar TT)** (primary)
- **Senapati Bapat Marg (Lower Parel to Elphinstone / Dadar)** (primary)
- **Rafi Ahmed Kidwai Marg (Sewri to Wadala)** (secondary)

METHODOLOGY & EXPOSURE CAVEAT:

Exposure figures represent geographic co-location within digital elevation sink buffers and BRIMSTOWAD catchments. They do not constitute verified flooding or structural damage. Actual inundation is subject to local stormwater drain maintenance, operational dewatering pumps, and real-time gate configurations.

MATHEMATICAL METHODOLOGY & INPUT CONDITIONS

SECTION E: FORMULA SPECIFICATION, TERRAIN INTEGRATION & AUDIT TRACE

1. MATHEMATICAL FORMULATION

HYDRODYNAMIC SCREENING

```
S_pond(t) = min(100, 0.35 * (P_excess(t) / 45 * 100) + 0.25 * D_topo + 0.20 * H_hotspot + 0.15 * d_drainage + 0.05 * S_accum(t))
```

Latex Representation: `S_pond(t) = min(100, 0.35 * (P_excess(t) / 45 * 100) + 0.25 * D_topo + 0.20 * H_hotspot + 0.15 * d_drainage + 0.05 * S_accum(t))`
Baseline Definition: Dry-weather baseline assuming 0 mm/h rain, 1.5m astronomical tide, 0% drainage blockage, and historical hotspot background. Delta percent represents: ((Score - Baseline) / max(1, Baseline)) * 100%.
Spatial Resolution: 24 Canonical BMC Administrative Wards + 250m Continuous Raster Grid • **Temporal Resolution:** Hourly increments from T+0h to T+6h

2. USER-CONTROLLED SCENARIO INPUTS & OBSERVATIONS

PARAMETER AUDIT

INPUT PARAMETER / VARIABLE	VALUE	UNIT	STATUS	SOURCE / NOTES
Scenario Precipitation Rate	45	mm/h	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Event Duration	3	hours	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Culvert & Nullah Siltation	15	%	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Temporal Rainfall Profile	0	profile	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Automatic Weather Station (AWS) Precipitation (Live Telemetry)	0	mm/h	ACTIVE	Open-Meteo High-Resolution AWS Feed
Apollo Bandar Astronomical Tide Level (Live Telemetry)	1.85	meters Chart Datum (CD)	ACTIVE	Mumbai Port Trust (MbPT) Tide Gauge Table

3. MISSING OR OFFLINE TELEMETRY FEEDS

HONEST DISCLOSURE

VARIABLE NAME	IMPACT ON SIMULATION MODEL	DETERMINISTIC FALLBACK USED
Real-time Drainage Sluice Gate SCADA	Outfall gate blockage % is user-configured rather than live sensor-driven.	`Scenario parameter default applied with explicit assumption note.`
Mithi River Ultrasonic Level Sensors	River corridor ponding modeled via static Copernicus elevation buffer.	`GLO-30 DEM 5.0m elevation contour threshold.`

RECOMMENDED ACTION DIRECTIVES & STATUTORY DISCLAIMER

SECTION F: INTER-AGENCY OPERATIONS & EXECUTIVE SIGN-OFF

1. DEPARTMENTAL OPERATIONAL ACTION DIRECTIVES

PRIORITY INTERVENTIONS

PHASE	DEPT	ACTION TITLE & DIRECTIVES	TARGET WARDS	PRIORITY
BEFORE_EVENT	SWD	**Pre-position High-Capacity Mobile Dewatering Pumps** Deploy auxiliary diesel dewatering pumps to documented depression basins in Ward G/North (Dadar / Dharavi / Mahim) and adjacent saucer basins prior to peak tidal window.	G/North, F/South, F/North	CRITICAL
DURING_MONITORING	ROADS	**Traffic Diversion Readiness at Key Underpasses** Stand by traffic police units at Hindmata, King Circle, and Kurla low-lying junctions for potential dynamic traffic diversion if waterlogging begins.	G/North, F/South	HIGH
BEFORE_EVENT	HEALTH	**Ground-Floor Flood Barrier Verification at Tertiary Hospitals** Ensure temporary flood barriers and sump pump backup generators are tested at KEM, Sion, and Nair hospital ground complexes.	F/South, F/North, E	HIGH
DURING_MONITORING	DMD	**Emergency Operations Centre (EOC) Real-Time Telemetry Watch** Activate dedicated situational display in MCGM Disaster Management Department EOC tracking AWS rain gauge rates and tidal curve.	G/North, F/South, F/North, E, L	CRITICAL
POST_EVENT	SWM	**Post-Event Trash Screen & Culvert Desilting Sweep** Conduct immediate desilting sweep of roadside catch-pits and storm water outfall screens following recession of runoff.	G/North, F/South, F/North, E, L	MEDIUM

2. AI GOVERNANCE & DATA PROVENANCE

AUDIT TRACE

AI Orchestration State: DETERMINISTIC_FALLBACK (138 ms)
Role 1 (Evidence Analyst): FAILED_AUTH (deterministic-rules-engine)
Role 2 (Impact Analyst): FALLBACK_DETERMINISTIC (deterministic-rules-engine)
Role 3 (Report Editor): FALLBACK_DETERMINISTIC (deterministic-rules-engine)

PREPARED BY:

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Signature & Municipal Seal

EXECUTIVE REVIEW & AUTHORISATION:

Director / Additional Municipal Commissioner
Disaster Management Department, MCGM Head Office

Authorisation Signature

STATUTORY DISCLAIMER & LIMITATIONS: This document constitutes a computerized multi-criteria hydrodynamic screening evaluation based on digital elevation modeling (Copernicus GLO-30 DEM) and BRIMSTOWAD storm drainage catchments. It is designed to assist administrative situational awareness and pre-positioning. It does NOT constitute an automated municipal proclamation, legal evacuation order, or public emergency dispatch. Official deployment of personnel, equipment, and public alarms requires formal sign-off by the Municipal Commissioner or designated Disaster Management Officer.